Parametric Survival Analysis

Stanford Heart Transplant Study

Group 6

Philomene CYUBAHIRO
Peter Onimisi ATTAIGU
Foumban Marah Khadij NOURIA
Yosamu MUHANZI
Zo Lalaina Andrianina ANDRIANANTENAINA

AIMS | African Institute for Mathematical Sciences RWANDA

March 22, 2026

Outline

1. Introduction
2. Data Summary
3. Data Visualization
4. Model Fitting Table
5. Best Model Selection
6. Life Function Plots
7. Interpretation
8. Conclusions

Introduction

Heart disease is one of the leading causes of death worldwide [2]. For patients with end-stage heart failure, cardiac transplantation remains the gold standard treatment, although survival outcomes vary widely between patients [1].

Objective

Fit and compare parametric survival models to the `stanford2` dataset, select the best model using AIC, and interpret the life functions in a clinical context.

Dataset: Stanford Heart Transplant Study

Dataset Overview

- Source: `stanford2` in R's `survival` package
- Stanford Heart Transplant Programme (1967–1980)

Table 1: Key Variables in the Dataset

Variable	Description
`time`	Follow-up time (days)
`status`	1 = death, 0 = censored
`age`	Age at transplant (years)
`t5`	T5 mismatch score
`id`	Patient ID

Table 2: Sample Summary Statistics

Statistic	Value
Total patients	184
Deaths (status = 1)	113
Censored (status = 0)	71

Descriptive Statistics

Table 3: Descriptive Statistics of Key Variables

Variable	N	Missing	Mean (SD)	Median	Min	Max
time	184	0	696.94 (802.47)	351.00	0.50	3695.00
age	184	0	41.09 (11.04)	44.00	12.00	64.00
t5	157	27	1.12 (0.58)	1.04	0.00	3.05

How long did they survive?

The average follow-up time after transplantation was about 697 days, roughly one year and ten months. But that average hides a painful truth: the middle value, the median, was just 351 days. Half of these 184 people survived less than a year after their transplant. Some held on for nearly 10 years (3,695 days which is the longest recorded). Others survived less than a single day.

Data Visualization

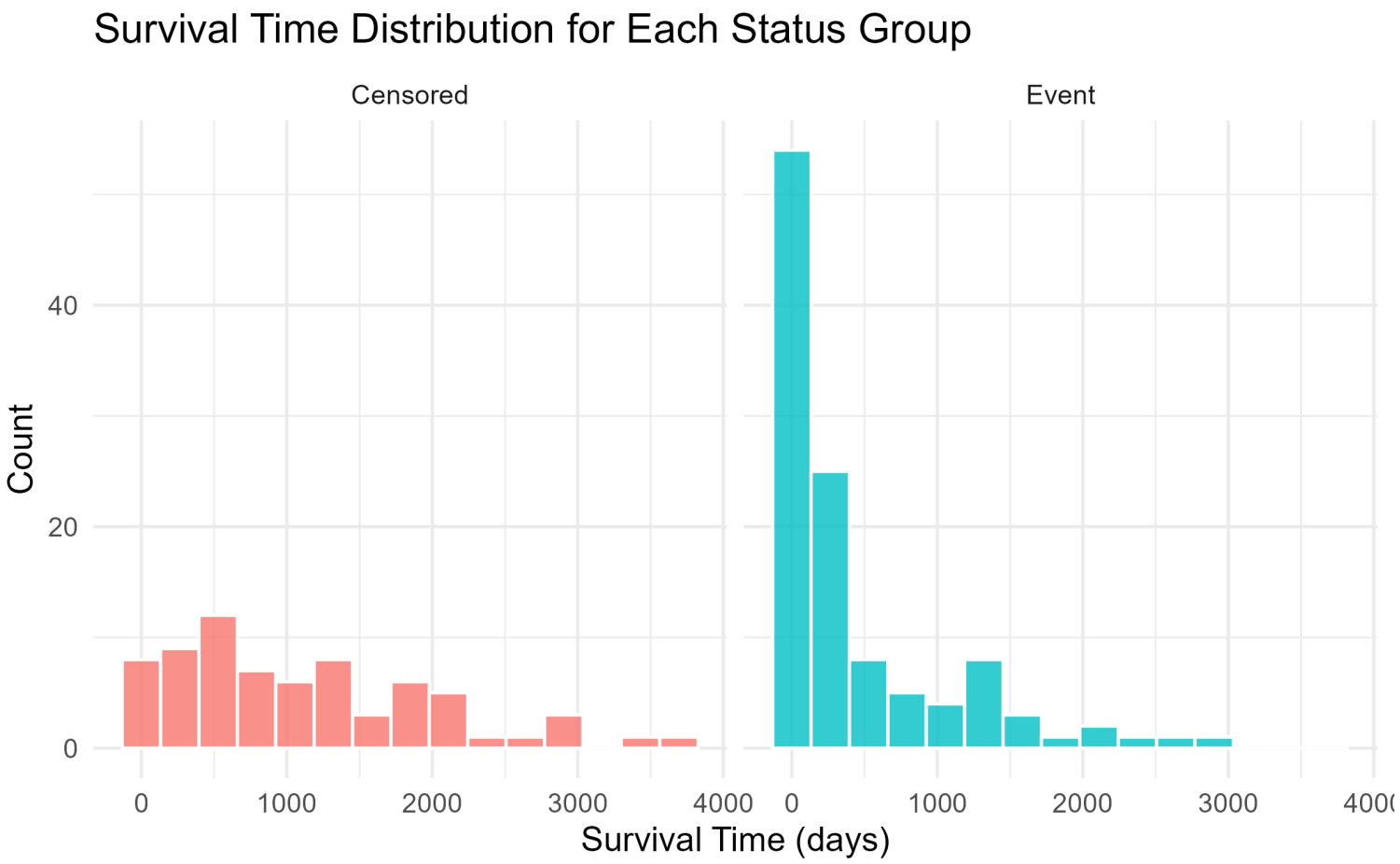

Figure 1: Survival Time Distribution

Data Visualization

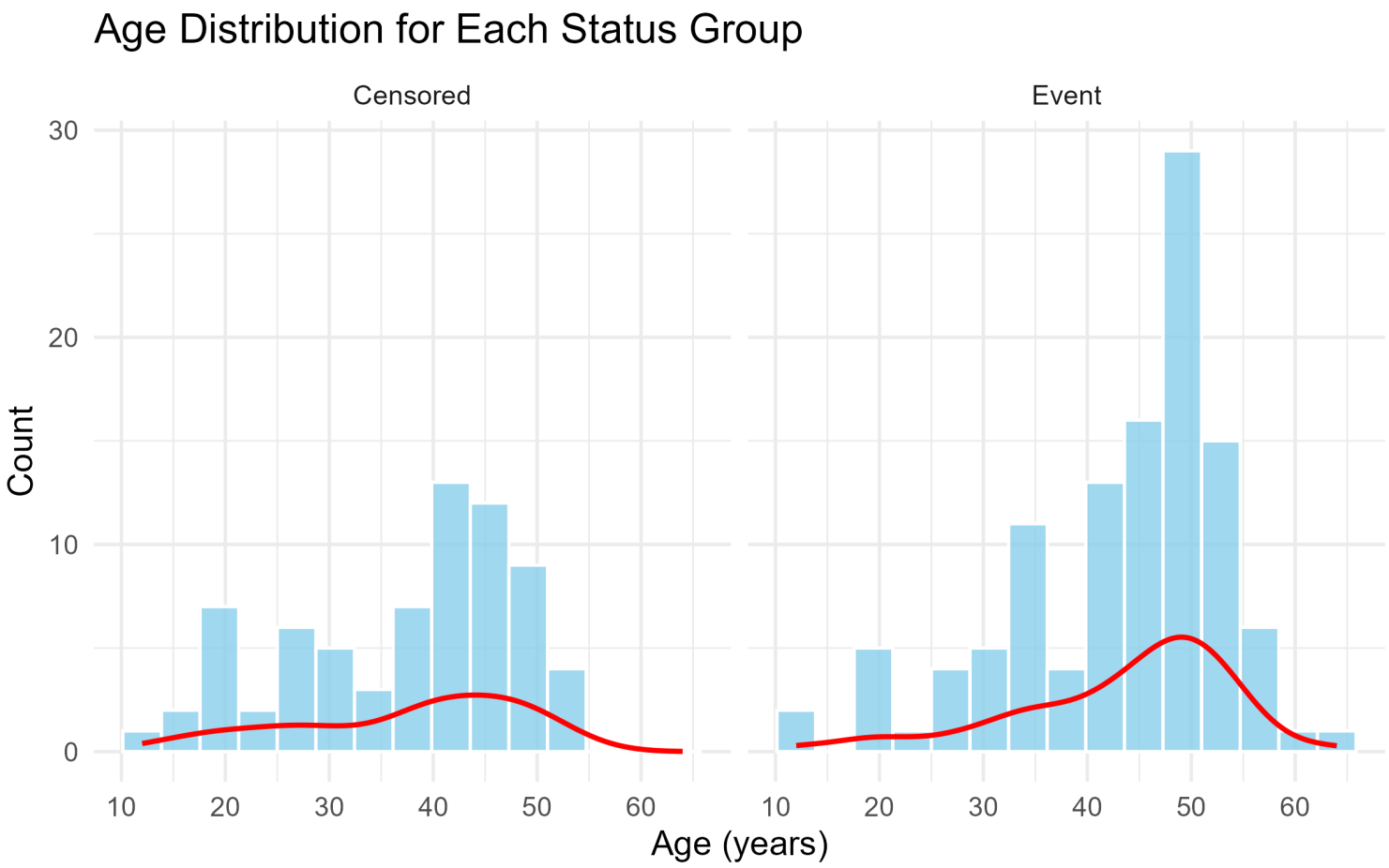

Figure 2: Age Distribution

Model Fitting and Comparison

Rank	Model	Log-Likelihood	AIC
1	**Log-Normal**	−868.806	**1741.611**
2	Generalised Gamma	−868.765	1743.530
3	Log-Logistic	−870.276	1744.553
4	Weibull	−871.752	1747.504
5	Gamma	−874.018	1752.037
6	Exponential	−907.870	1817.741

Table 4: AIC comparison of all fitted parametric models

Best Model

The **Log-Normal** model has the lowest AIC = **1741.611** and is selected as the best-fitting model.

MLE Estimates – Log-Normal Model

Parameter	Estimate	SE	95% CI Lower	95% CI Upper
$\hat{\mu}$ (meanlog)	6.280	0.202	5.885	6.675
$\hat{\sigma}$ (sdlog)	2.431	0.172	2.116	2.792

Table 5: MLE estimates

$\hat{\mu} = 6.280$

Mean of log survival times.
Median survival $= e^{6.280} \approx$ **534 days**

$\hat{\sigma} = 2.431$

Spread of log survival times.
Large value $\Rightarrow$ **high variability** in patient outcomes.

Mean Survival Time

$\approx$ **10,248 days**

Variance

$\approx$ **3.86×10^{10} days^2**

KM Curve vs Log-Normal Model

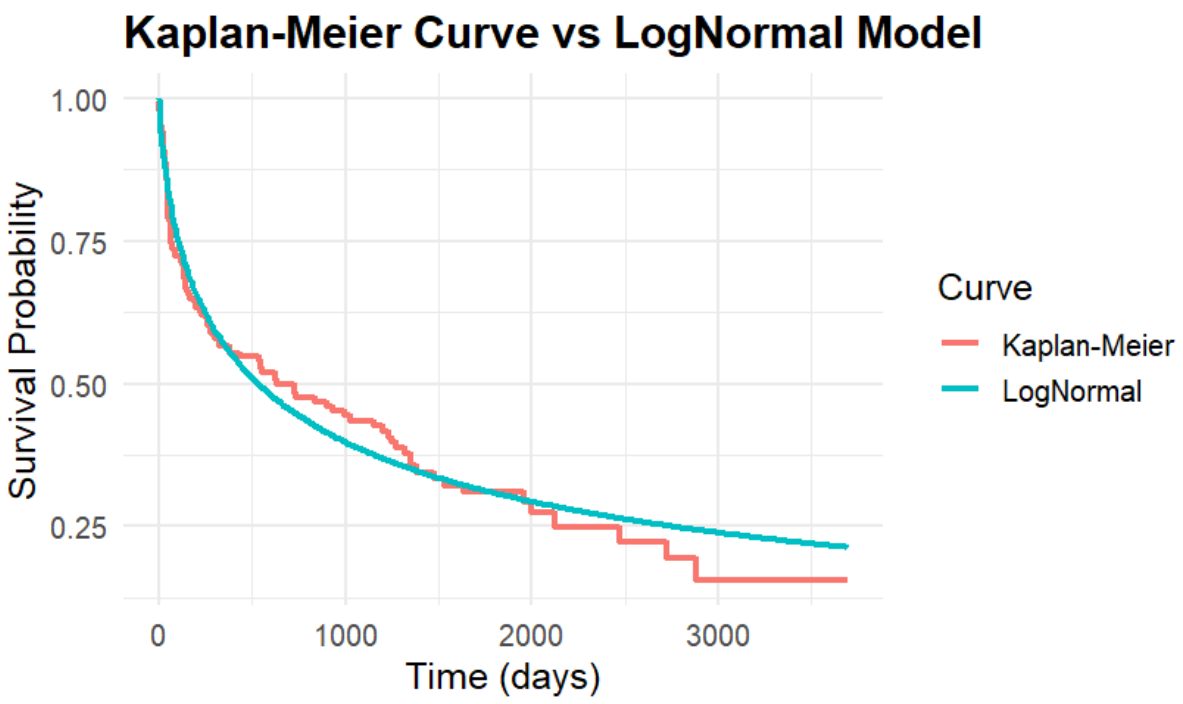

Figure 3: KM Curve vs Log-Normal Model

Interpretation: The Kaplan-Meier curve gives us the raw, observed survival of the 184 patients, calculated directly from the data with no assumptions.

The model we built closely mirrors what actually happened, giving us confidence that its predictions about survival, risk, and time are trustworthy, not just theoretical.

Life Function Plots – Log-Normal Model

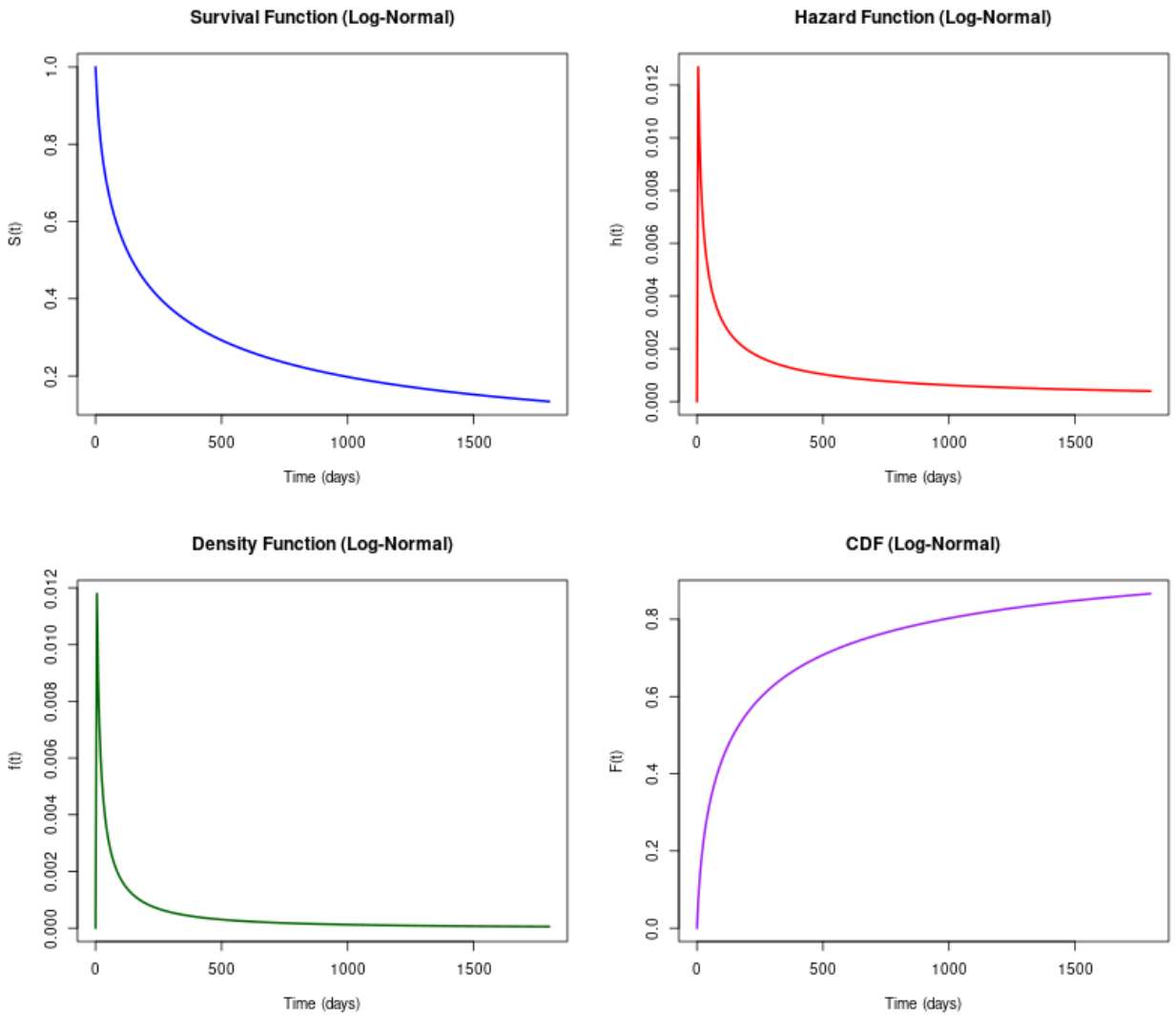

Figure 4: S(t), h(t), f(t) and F(t) under the Log-Normal model

Interpretation of Life Functions

- **S(t) – Survival Function:** By day 3,500 (nearly ten years later), only about 20% of patients are still alive. The median survival is approximately 534 days. This means that by around day 534, roughly one year and five months after surgery, half of all patients had already died
- **h(t) – Hazard Function:** Rises sharply in the early days then gradually declines, a non-monotone pattern. Risk of death is highest immediately after transplant surgery, then falls for long-term survivors.
- **f(t) – Density Function:** Right-skewed distribution. Most deaths occur in the early post-transplant period, with a long tail of survivors.
- **F(t) – Cumulative Distribution:** By the time the study concluded, approximately 8 in every 10 transplant patients had died(80%). The remaining 20% represented by the 71 censored patients in our data.

Conclusions & Clinical Insights

Conclusions

- The **Log-Normal** model was selected as the best fit (AIC $= 1741.611$) from 6 candidate models
- MLE estimates: $\hat{\mu} = 6.280$, $\hat{\sigma} = 2.431$, giving a **median survival of** ≈ 534 **days** and **mean survival of** $\approx 10{,}248$ **days**

Clinical Insights

- Healthcare systems should ensure intensive monitoring and support in the first 12 to 18 months after surgery.
- Health systems must invest in personalised follow-up plans because of the large variation in survival times, from 12 hours to nearly 10 years.
- Health literacy programs should be strengthened.

Thank You!

Questions & Discussion

GitHub Repository: Survival Analysis Heart Transplant Project

References

International Society for Heart and Lung Transplantation.
Ishlt transplant registry quarterly reports, 2022.
Accessed: 2026-03-21.

World Health Organization.
Cardiovascular diseases (cvds), 2023.
Accessed: 2026-03-21.